# DVR FILL-IN CRITERION FOR SEPARATEDNESS

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 2840-2860 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 2862-2906, the complete author proof (43 body lines).

*Editorial note.* Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Queue index 29 of queue_stacks-stacks-more-morphisms.json (pb=1909, stmt 2840-2860, proof 2862-2906).

*Editorial note.* stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

*Editorial note.* Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-check-separated-dvr' also exists in stacks-spaces-cohomology — a different proposition in a different chapter file.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

*Source reference.* Original lemma 0ARK: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2906–2906. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0CLA: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2906–2906. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0CLE: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2906–2906. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0E8L: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2906–2906. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 06R6: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2906–2906. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 04XS: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2906–2906. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 04XD: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2906–2906. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. Assume*

(1) *$\mathcal{Y}$ is locally Noetherian,*
(2) *$f$ is locally of finite type and quasi-separated,*
(3) *for every commutative diagram*

$$\begin{array}{ccc} \mathrm{Spec}(K) & \xrightarrow{x} & \mathcal{X} \\ {\scriptstyle j}\downarrow & \nearrow & \downarrow{\scriptstyle f} \\ \mathrm{Spec}(A) & \xrightarrow{y} & \mathcal{Y} \end{array}$$

*where $A$ is a discrete valuation ring and $K$ its fraction field and any 2-arrow $\gamma : y \circ j \to f \circ x$ the category of dotted arrows (Morphisms of Stacks, Definition 0CLA) is either empty or a setoid with exactly one isomorphism class.*

*Then $f$ is separated.*

*Proof.* To prove that $f$ is separated we have to show that $\Delta : \mathcal{X} \to \mathcal{X} \times_{\mathcal{Y}} \mathcal{X}$ is proper. We already know that $\Delta$ is representable by algebraic spaces, locally of finite type (Morphisms of Stacks, Lemma 04XS) and quasi-compact and quasi-separated (by definition of $f$ being quasi-separated). Choose a scheme $U$ and a surjective smooth morphism $U \to \mathcal{X} \times_{\mathcal{Y}} \mathcal{X}$. Set

$$V = \mathcal{X} \times_{\Delta, \mathcal{X} \times_{\mathcal{Y}} \mathcal{X}} U$$

It suffices to show that the morphism of algebraic spaces $V \to U$ is proper (Properties of Stacks, Lemma 04XD). Observe that $U$ is locally Noetherian (use Morphisms of Stacks, Lemma 06R6 and the fact that $U \to \mathcal{Y}$ is locally of finite type) and $V \to U$ is of finite type and quasi-separated (as the base change of $\Delta$ and properties of $\Delta$ listed above). Applying Cohomology of Spaces, Lemma 0ARK it suffices to show: Given a commutative diagram

$$\begin{array}{ccccc} \mathrm{Spec}(K) & \xrightarrow{v} & V & \longrightarrow & \mathcal{X} \\ {\scriptstyle j}\downarrow & \nearrow & {\scriptstyle g}\downarrow & \nearrow & \downarrow{\scriptstyle \Delta} \\ \mathrm{Spec}(A) & \xrightarrow{u} & U & \longrightarrow & \mathcal{X} \times_{\mathcal{Y}} \mathcal{X} \end{array}$$

where $A$ is a discrete valuation ring and $K$ its fraction field, there is a unique dashed arrow making the diagram commute. By Morphisms of Stacks, Lemma 0CLE the categories of dashed and dotted arrows are equivalent. Assumption (3)

implies there is a unique dotted arrow up to isomorphism, see Morphisms of Stacks, Lemma 0E8L. We conclude there is a unique dashed arrow as desired. □